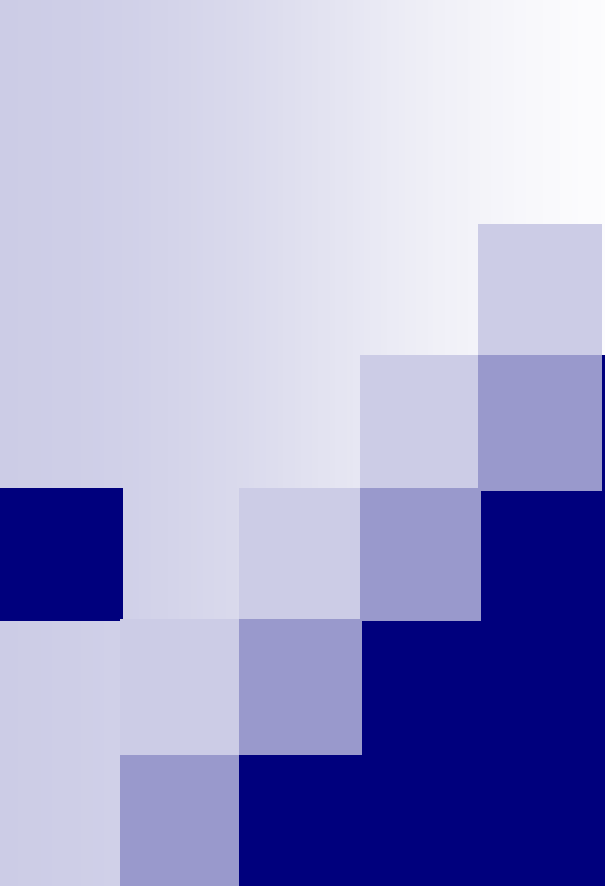




Software Quality Assurance



Lecture # 5

Project Management Approaches and High Software Quality - 1

- Use of automated project estimation methods
- Use of automated project planning methods
- Use of early and automated estimates of software defect potentials
- Use of early and automated estimates of software defect removal efficiency
- Formal risk-analysis

Project Management Approaches and High Software Quality - 2

- Provision of adequate time for pre-test inspections
- Historical quality data from similar projects available
- Milestone tracking automated and thorough
- Defect tracking automated and thorough
- Management focus concentrated on achieving excellent results

Project Management Approaches and Poor Software Quality

- Exact opposite of the project management approaches correlating with high software quality
- In the previous lecture we talked about a software quality assurance (or SQA) group. Let's now see what is an SQA group, why we need it, and what are its activities

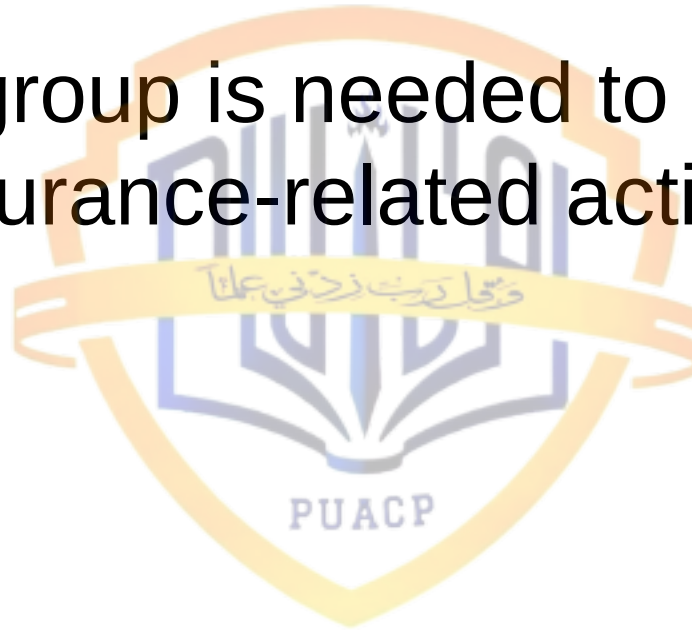
SQA Group - 1

- Every company, which wants to establish a reputation for producing high quality software, must establish a Software Quality Assurance (SQA) Group within the company
- This groups must be funded properly and management must pay attention to the reports and presentations made by this group

SQA Group - 2

- The SQA group report directly to the line-management and not to the project management
- The personnel of the SQA group must work with the project management team, and vice versa to produce high quality software for the company – which is the ultimate goal

- The SQA group is needed to monitor the quality assurance-related activities in a company



SQA Group's Activities - 1

- Preparation of an SQA plan for a project
- Participation in the development of the project's software process description
- Review of software engineering activities to verify compliance with the defined software process

SQA Group's Activities - 2

- Audit of designed software work products to verify compliance with those defined as part of the software process



SQA Group's Activities - 3

- Ensure that deviations in software work and work products are documented and handled according to a documented procedure
- Record any noncompliance and reports to senior management

SQA Plan - 1

- Evaluations to be performed
- Audits and reviews to be performed
- Standards that are applicable to the project
- Procedures for error reporting and tracking

SQA Plan - 2

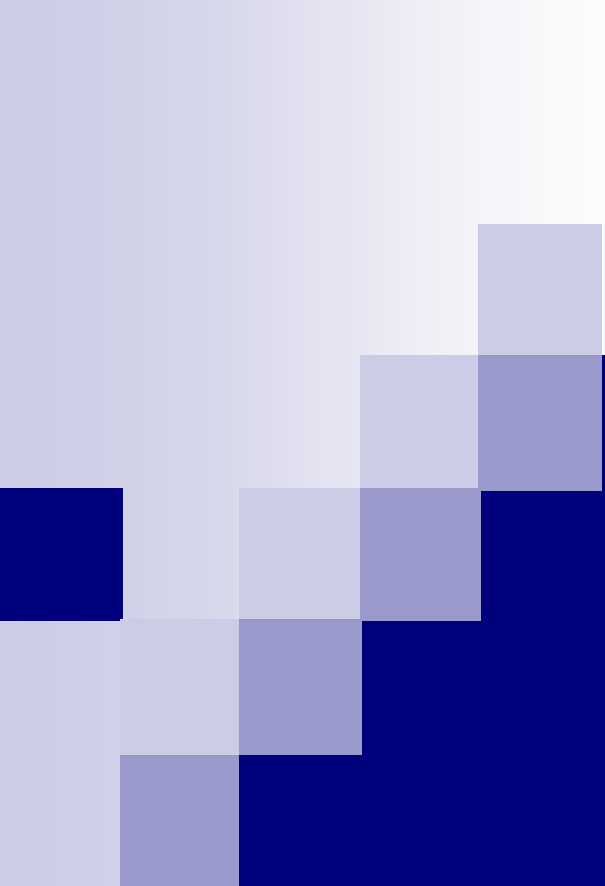
- Documents to be produced by the SQA group
- Amount of feedback provided to the software project team
- (we'll discuss SQA Plan in detail later in the course)

Software Quality Personnel

- Unfortunately are under-paid
- Usually are let go first in times of crisis
- “Top-gun” SQA personnel and managers with proven track record are in high demand from companies that have active QA programs

Costs of Software Quality - 1

- Defects prevention costs
- User satisfaction optimization costs
- Data quality defect prevention costs
- Data quality defect removal costs
- Quality awareness/training costs
- Non-test defect removal costs
- Testing defect removal costs



Costs of Software Quality

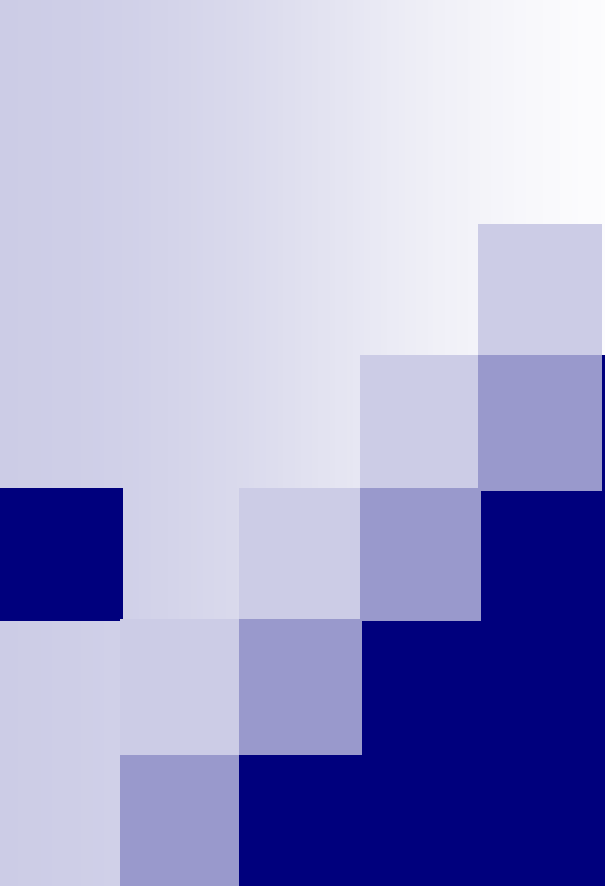


Costs of Software Quality - 2

- Post-release customer support costs
- Litigation and damage award costs
- Quality savings from reduced scrap/rework
- Quality savings from reduced user downtime
- Quality value from reduced time-to-market intervals

Costs of Software Quality - 3

- Quality value from enhanced competitiveness
- Quality value from enhanced employee morale
- Quality return on investment



Economics of Software Quality



Economics of Software Quality - 1

- High quality software applications have shorter development schedules than low quality applications because they do not get hung up in integration and testing due to excessive defect levels

Economics of Software Quality - 2

- High quality software applications have lower development and maintenance costs than low quality applications. This is because the cumulative costs of finding and fixing bugs is often the major cost driver for software projects

Economics of Software Quality - 3

- High quality software applications have better reliability levels and longer mean times to failure than low quality applications
- High quality commercial software packages have larger market shares than low quality commercial software packages

Economics of Software Quality - 4

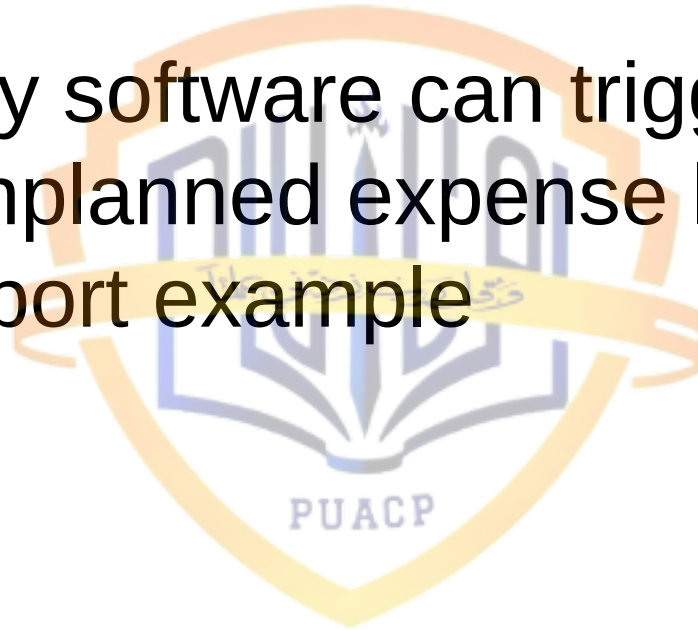
- High quality software achieves better user-satisfaction ratings than low quality software
- High quality software projects score better on employee morale surveys than do low quality software projects

Economics of Software Quality - 5

- High quality software produced under contract or an outsource agreement has a much lower probability of ending up in court for breach of contract or malpractice litigation than low quality software
- High quality software benefits or augments the performance levels of users, while poor quality tends to degrade worker performance

Economics of Software Quality - 6

- Poor quality software can trigger truly massive unplanned expense levels.
Denver airport example



References

- Software Quality: Analysis and Guidelines for Success by Capers Jones
- Customer-Oriented Software Quality Assurance by Frank Ginac
- A Practitioner's Approach to Software Engineering by Roger Pressman